

PROJECT REPORT

RetailVision AI

Real Time Video Analytics for Retail

Submitted in Partial Fulfillment of the Requirements for the Successful Completion of the Internship Program

Submitted By

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CHAPTER 1

INTRODUCTION

1.1 Introduction

The retail industry increasingly relies on data-driven decision-making to improve customer experience, optimize store layouts, and enhance operational efficiency. While modern retail stores are equipped with surveillance cameras, the recorded video is primarily used for security purposes, leaving valuable customer behavior information largely unexplored.

Recent advancements in Artificial Intelligence (AI) and Computer Vision have enabled surveillance footage to be analyzed automatically, providing meaningful insights into customer movement, shopping patterns, and store occupancy. Technologies such as object detection and multi-object tracking allow businesses to understand how customers interact with different areas of a store without requiring manual observation.

This project, **RetailVision AI**, presents an AI-powered retail video analytics system capable of detecting customers, tracking their movement across video frames, estimating dwell time within predefined zones, and generating customer movement heatmaps. The processed information is presented through an interactive Streamlit dashboard, enabling users to visualize key retail analytics in a simple and intuitive manner.

The system integrates **YOLOv8** for real-time person detection, **DeepSORT** for multi-object tracking, **OpenCV** for video processing, and **Streamlit** for dashboard visualization. Together, these technologies transform ordinary retail surveillance footage into actionable business intelligence that can assist store managers in understanding customer behavior and making informed operational decisions.

The project demonstrates the practical application of computer vision in retail environments while showcasing how artificial intelligence can automate customer analytics, reduce manual effort, and support data-driven retail management.

1.2 Problem Statement

Retail stores generate a significant amount of surveillance footage every day. However, most of this video data is used only for security and is rarely analyzed to understand customer behavior. Manually monitoring customer movement, estimating dwell time, and identifying high-traffic areas is time-consuming, inefficient, and prone to human error.

The objective of this project is to develop an AI-powered retail video analytics system capable of automatically detecting customers, tracking their movement, analyzing customer activity within predefined store zones, and presenting meaningful business insights through an interactive dashboard.

1.3 Objectives

The primary objectives of the project are:

- To detect customers from retail surveillance videos using YOLOv8.
 - To perform multi-object tracking using the DeepSORT algorithm.
 - To calculate the number of unique customers and peak store occupancy.
 - To analyze customer visits across predefined store zones.
 - To estimate customer dwell time within each zone.
 - To generate customer movement heatmaps for visual analysis.
 - To develop an interactive Streamlit dashboard for presenting retail analytics.
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1.4 Scope of the Project

The project focuses on analyzing recorded retail surveillance videos to generate customer behavior analytics. It performs person detection, customer tracking, zone-wise analytics, dwell time estimation, and heatmap generation using computer vision techniques.

The developed system demonstrates how AI can transform surveillance footage into actionable business intelligence. Although the current implementation processes recorded videos, the overall pipeline can be extended to support live CCTV feeds, cloud deployment, and advanced retail analytics in future implementations.

1.5 Organization of the Report

The report is organized into six chapters.

- **Chapter 1** introduces the project, defines the problem statement, objectives, and project scope.
- **Chapter 2** reviews the existing literature and related technologies.
- **Chapter 3** presents the proposed system architecture, workflow, and technology stack.
- **Chapter 4** explains the implementation methodology, including customer detection, tracking, zone analytics, dwell time estimation, heatmap generation, and dashboard development.
- **Chapter 5** discusses the experimental results and dashboard outputs.
- **Chapter 6** conclusion, future scope and references.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Computer vision has become one of the most significant applications of Artificial Intelligence, enabling machines to understand and analyze visual information from images and videos. In the retail industry, computer vision techniques are increasingly being used to automate customer analytics, monitor store activity, improve operational efficiency, and support business decision-making.

This chapter reviews the key technologies and concepts used in the development of the proposed retail analytics system.

2.2 Retail Video Analytics

Retail video analytics refers to the process of extracting meaningful business information from surveillance footage using computer vision and artificial intelligence. Instead of using CCTV cameras solely for security purposes, retailers can analyze customer movement, occupancy, dwell time, and shopping behavior to improve store operations and customer experience.

Modern retail analytics systems help store managers identify high-traffic areas, evaluate customer engagement, optimize product placement, and improve workforce planning.

2.3 YOLOv8 for Object Detection

YOLO (You Only Look Once) is a real-time object detection algorithm designed to identify objects accurately while maintaining high processing speed. YOLOv8, developed by Ultralytics, improves detection accuracy and efficiency compared to earlier versions, making it suitable for real-time applications.

In this project, YOLOv8 is used to detect customers in each video frame. The detected bounding boxes serve as the input for the tracking module, enabling continuous monitoring of customer movement throughout the store.

2.4 DeepSORT for Multi-Object Tracking

DeepSORT (Deep Simple Online and Realtime Tracking) is a multi-object tracking algorithm that assigns unique identities to detected objects and maintains those identities across consecutive video frames.

Unlike simple tracking methods, DeepSORT combines motion prediction and appearance information, allowing customers to retain consistent IDs even when temporary occlusions occur.

In this project, DeepSORT is responsible for assigning unique customer IDs, preventing duplicate counting, and enabling dwell time estimation.

2.5 OpenCV

OpenCV (Open Source Computer Vision Library) is a widely used computer vision library that provides tools for image and video processing.

Within this project, OpenCV is used to read video frames, draw bounding boxes, display tracking information, define analytical zones, generate heatmaps, and save the processed output video.

2.6 Streamlit

Streamlit is an open-source Python framework used for building interactive web applications with minimal code.

It enables rapid development of dashboards for visualizing analytical results. In this project, Streamlit presents key performance indicators, processed video output, zone analytics, dwell time reports, and customer movement heatmaps through an intuitive interface.

2.7 Summary

The technologies discussed in this chapter form the foundation of the proposed retail analytics system. YOLOv8 performs customer detection, DeepSORT enables reliable multi-object tracking, OpenCV handles video processing, and Streamlit provides an interactive platform for visualizing business insights. Together, these technologies create an end-to-end retail video analytics solution capable of transforming surveillance footage into actionable information.

PROPOSED SYSTEM

3.1 Introduction

The proposed system is an AI-powered retail video analytics platform designed to transform surveillance footage into meaningful business insights. The system integrates computer vision techniques with interactive visualization to automatically detect customers, track their movement, analyze customer behavior, and present the generated analytics through a user-friendly dashboard.

The complete workflow consists of video processing, customer detection, multi-object tracking, analytics generation, and dashboard visualization.

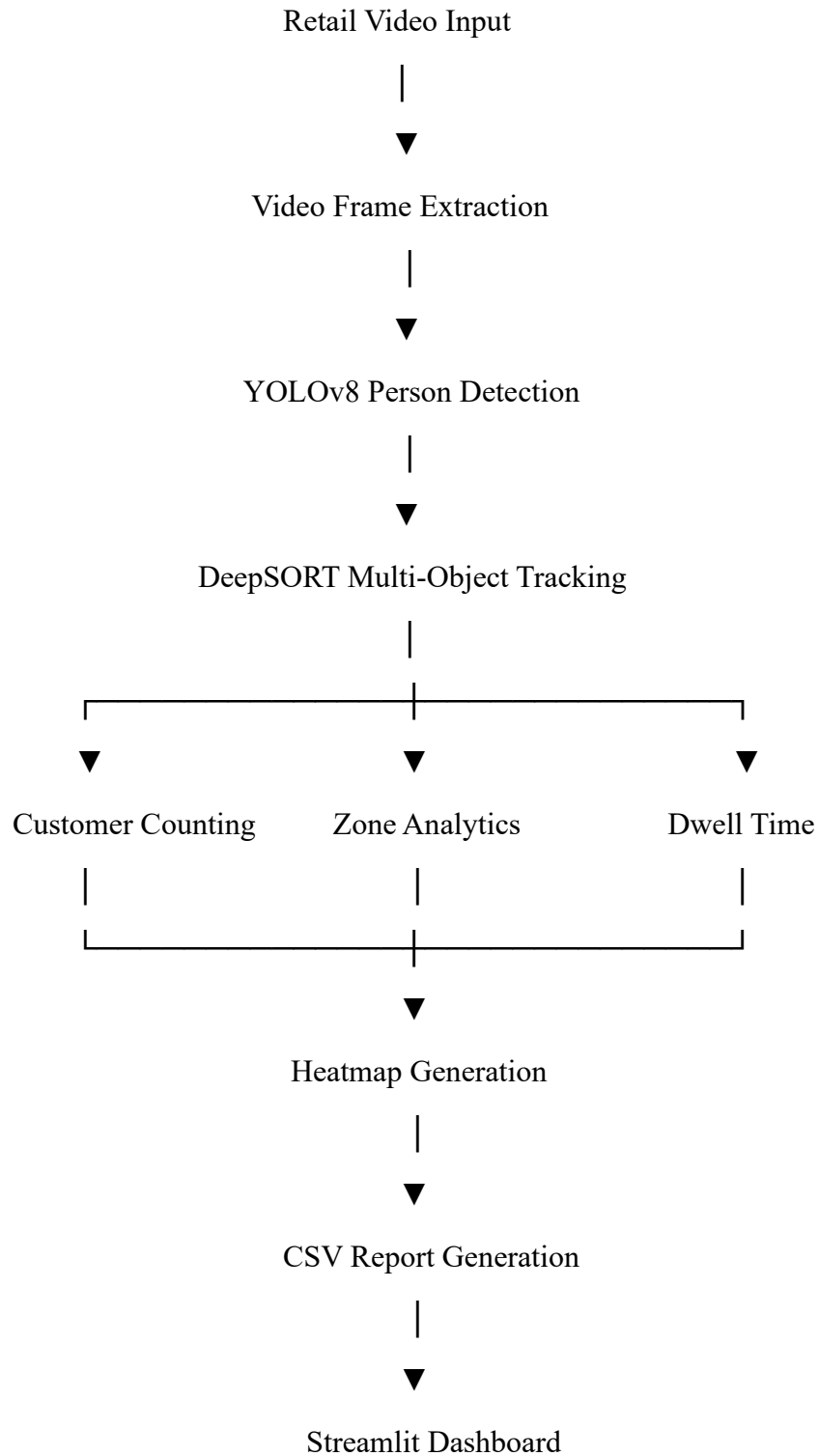
3.2 System Workflow

The execution of the proposed system follows these steps:

1. Read the retail surveillance video.
2. Extract video frames using OpenCV.
3. Detect customers using the YOLOv8 model.
4. Assign unique IDs to detected customers using DeepSORT.
5. Track customer movement across consecutive frames.
6. Identify customer visits within predefined store zones.
7. Calculate customer dwell time for each zone.
8. Generate a customer movement heatmap.
9. Store analytical results in CSV files.
10. Display all analytics using the Streamlit dashboard.

3.3 System Architecture

The overall architecture of the proposed system is illustrated below.



3.4 System Modules

3.4.1 Video Processing Module

This module reads the retail surveillance video using OpenCV and extracts individual frames for further analysis. It also handles video writing after customer tracking annotations have been added.

3.4.2 Customer Detection Module

The YOLOv8 model detects customers in every frame by identifying the "person" class. Bounding boxes are generated for each detected customer, providing the input for the tracking module.

3.4.3 Customer Tracking Module

DeepSORT assigns a unique tracking ID to every detected customer. The tracking IDs remain consistent across multiple frames, enabling accurate customer counting and movement analysis.

3.4.4 Zone Analytics Module

Predefined regions inside the retail store are monitored to determine customer visits. Each time a tracked customer enters a zone, the visit count is updated, allowing the system to identify high-traffic areas.

3.4.5 Dwell Time Analysis Module

The dwell time module calculates the duration each customer spends inside a predefined zone. The calculated values help estimate customer engagement with different areas of the store.

3.4.6 Heatmap Generation Module

Customer movement coordinates collected during tracking are accumulated to generate a heatmap. The heatmap visually represents customer activity density and highlights frequently visited areas.

3.4.7 Dashboard Module

The Streamlit dashboard displays processed videos, key performance indicators, heatmaps, zone analytics, and dwell time reports through an interactive graphical interface. It provides users with an intuitive summary of all generated analytics.

3.5 Advantages of the Proposed System

- Automated customer detection and tracking.
 - Accurate customer counting without manual observation.
 - Real-time zone-wise customer analytics.
 - Customer dwell time estimation.
 - Interactive dashboard for business visualization.
 - Scalable design that can be extended for live CCTV deployment.
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3.6 Summary

The proposed system integrates object detection, multi-object tracking, video processing, analytics generation, and dashboard visualization into a unified pipeline. The modular architecture enables efficient customer behavior analysis while providing meaningful business insights for retail environments.

CHAPTER 4

IMPLEMENTATION

4.1 Introduction

This chapter describes the implementation of the proposed RetailVision AI system. The project was developed using Python and integrates multiple computer vision and data processing libraries to analyze retail surveillance footage. The implementation consists of customer detection, multi-object tracking, zone analytics, dwell time estimation, heatmap generation, and dashboard visualization.

4.2 Development Environment

Component	Specification
Programming Language	Python
IDE	Visual Studio Code
Dashboard Framework	Streamlit
Computer Vision Library	OpenCV
Object Detection	YOLOv8
Object Tracking	DeepSORT
Data Processing	Pandas
Operating System	Windows 11 (64-bit)

4.3 Video Processing

The input retail surveillance video is processed frame by frame using OpenCV. Each frame is resized and passed to the object detection model for identifying customers. The processed frames are annotated with bounding boxes, tracking IDs, and zone boundaries before being written to the output video.

4.4 Customer Detection

YOLOv8 is used to detect customers in each video frame. Only the **person** class is considered for analysis. The model generates bounding boxes around detected customers, which are then passed to the tracking module.

The detection module provides:

- Customer location
 - Bounding box coordinates
 - Detection confidence
-

4.5 Multi-Object Tracking

DeepSORT assigns a unique identity to each detected customer. The assigned ID remains consistent across consecutive frames, enabling continuous customer tracking.

This tracking information is used to:

- Count unique customers
 - Estimate peak occupancy
 - Monitor customer movement
 - Prevent duplicate counting
-

4.6 Zone Analytics

The retail store is divided into predefined analytical zones.

Whenever a tracked customer enters a zone, the visit counter for that region is updated. This allows the system to identify high-traffic locations inside the store.

The generated zone analytics are exported to a CSV file and displayed on the Streamlit dashboard.

4.7 Dwell Time Estimation

The system records the time when a customer enters and exits a predefined zone.

The difference between the entry and exit timestamps is used to calculate customer dwell time.

This metric helps estimate customer engagement within different store sections.

4.8 Heatmap Generation

During tracking, the center point of each detected customer is recorded.

These coordinates are accumulated over the entire video to generate a movement heatmap.

The resulting heatmap highlights areas with high customer activity, providing useful insights into store traffic patterns.

4.9 Streamlit Dashboard

A Streamlit dashboard was developed to visualize the generated analytics.

The dashboard displays:

- Processed retail video
- Key Performance Indicators (KPIs)
- Customer movement heatmap
- Zone analytics
- Customer dwell time analysis

The dashboard provides an intuitive interface for interpreting retail analytics without requiring technical knowledge.

4.10 Output Files

The implementation generates the following output files:

File	Description
retail_analytics_output.mp4	Processed video with tracking annotations
retail_heatmap_overlay.png	Customer movement heatmap
summary.csv	Summary metrics
zone_analytics.csv	Zone-wise visit statistics
dwell_time.csv	Customer dwell time information

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents the results obtained from the implementation of the proposed RetailVision AI system. The developed application successfully analyzes retail surveillance videos by detecting customers, tracking their movement, generating analytical insights, and displaying the results through an interactive dashboard.

5.2 Customer Detection and Tracking

The YOLOv8 model successfully detected customers in the retail surveillance video. DeepSORT assigned unique IDs to each detected customer, enabling continuous tracking across multiple video frames. This approach ensured accurate customer counting while minimizing duplicate detections.

5.3 Dashboard Visualization

The developed Streamlit dashboard provides a user-friendly interface for visualizing the generated retail analytics. It displays processed video output, key performance indicators, customer movement heatmaps, zone analytics, and dwell time information in a single platform, making the analytical results easy to interpret.

5.4 Customer Movement Heatmap

The heatmap generated by the system represents customer movement patterns throughout the store. Areas with higher customer activity appear with greater intensity, allowing users to identify frequently visited regions and understand overall customer traffic distribution.

5.5 Zone Analytics

The predefined analytical zones successfully recorded customer visits during video processing. The generated zone analytics help identify high-traffic and low-traffic areas within the retail environment, providing useful information for store layout optimization and operational planning.

5.6 Customer Dwell Time

The system successfully calculated the time spent by customers within predefined zones. Dwell time analysis provides insights into customer engagement and helps evaluate how different sections of the store attract customer attention.

5.7 Discussion

The implemented system successfully demonstrates the application of computer vision techniques in retail analytics. By integrating customer detection, tracking, zone analysis, dwell time estimation, and heatmap generation, the system converts surveillance footage into meaningful business insights. The interactive dashboard further improves usability by presenting all analytical information in a structured and accessible manner.

5.8 Summary

The results demonstrate that the proposed RetailVision AI system effectively performs customer detection, tracking, analytics generation, and dashboard visualization. The generated outputs validate the practical applicability of the system for retail customer behavior analysis and operational decision-making.

CHAPTER 6

CONCLUSION

6.1 Conclusion

RetailVision AI demonstrates the practical application of Artificial Intelligence and Computer Vision in retail analytics. The developed system successfully detects customers, tracks their movement, analyzes customer activity within predefined store zones, estimates dwell time, and generates customer movement heatmaps from retail surveillance videos.

The integration of YOLOv8 for customer detection, DeepSORT for multi-object tracking, OpenCV for video processing, and Streamlit for dashboard visualization resulted in a complete end-to-end retail analytics solution. The generated analytics enable users to understand customer behavior, identify high-traffic areas, and obtain meaningful business insights without manual observation.

The project successfully achieved its objectives by automating retail video analysis and presenting the results through an interactive dashboard. It demonstrates how computer vision techniques can transform conventional surveillance systems into intelligent decision-support tools for retail management.

6.2 Future Scope

Although the proposed system achieves its intended objectives, several enhancements can be incorporated in future versions:

- Support real-time analysis using live CCTV camera feeds.
- Integrate cloud-based storage and remote dashboard access.
- Implement customer demographic analysis such as age and gender estimation.
- Develop customer trajectory and shopping path analysis.
- Generate automated business reports and trend analysis.
- Extend the system to support multiple camera feeds simultaneously.
- Integrate alert systems for overcrowding and unusual customer activity.

The modular design of the proposed system makes these enhancements feasible while maintaining the existing architecture.

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